Batterjee Medical College
*For Science and Technology*

Psychiatry · Clinical Lecture

# Psychotropic Medication

The four workhorse classes — antidepressants, anxiolytics/sedatives, antipsychotics and mood stabilisers: how they work, how to choose them, and their side-effects and emergencies.

Name: Dr. Hussain M. Khrad, MBBS, ABNP, ABAM, MA
Designation: Consultant Psychiatrist, KFSHRC-Jeddah · Psychiatry Residency Program Director (SCFHS)
Department: Department of Psychiatry

# Learning Objectives (1 of 2)



**Mapped to the BMC Knowledge / Skill / Value framework. Prerequisite: the disorder lectures — mood, anxiety and psychotic disorders.**

## Knowledge & understanding

- Name the four workhorse classes and how they act
- State the common and serious side-effects of each class
- Recognise serotonin syndrome, NMS and lithium toxicity

## Skills

- Choose an antidepressant by side-effect profile and overdose safety
- Separate serotonin syndrome from NMS at the bedside
- Reconcile psychotropics on admission and never stop them abruptly

## Values & attitudes

- Treat every prescription as a trade-off between efficacy and harm
- Ask directly about sexual and metabolic effects
- Consent properly: say what the drug costs as well as what it does

# Learning Objectives (2 of 2)



## Knowledge & understanding

- Know the EPS timeline and the treatment of each syndrome
- State lithium's range, monitoring and interactions
- Outline the black-box warnings that shape prescribing

## Skills

- Spot the everyday drugs that collide with a psychotropic
- Order the right baseline before the first dose, and monitor after
- Act on fever or sore throat in a patient on clozapine, the same day

## Values & attitudes

- Never give a drug lethal in overdose to a patient at risk
- Take physical health as seriously as mental health
- Involve the patient in the choice — the strongest adherence intervention

*Cross-cutting theme: every psychotropic is a balance of efficacy vs. side-effects. Matching drug to patient is the core skill, and recognising the emergencies — serotonin syndrome, NMS, lithium toxicity — can save a life.*

BMC
Batterjee Medical College

PART 1

# Antidepressants

Classes and principles · SSRIs · SNRIs · TCAs · MAOIs · newer agents · serotonin syndrome · discontinuation · pregnancy.

# Antidepressant classes & general principles

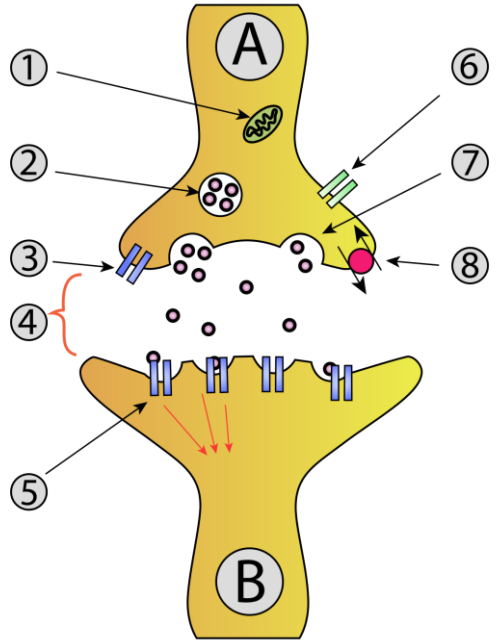
## Similar efficacy across classes — they differ in safety and side-effects.

### The major categories

- SSRIs
- SNRIs
- Heterocyclics — TCAs and tetracyclics
- MAOIs
- Miscellaneous — bupropion, mirtazapine, trazodone, newer agents

### Principles

- ~60–70% respond to an antidepressant
- Full benefit at 4–6 weeks on a given dose
- Start low, titrate slowly in the side-effect-prone




*Chemical synapse — 8 is the reuptake pump. Mouagip/Dake, Wikimedia Commons, CC BY-SA 3.0*

*Every class acts on the synapse above — the differences are in which transporters and receptors they touch, and therefore in what they do to the patient.*

# Choosing an antidepressant



**No single antidepressant is clearly most effective — match the drug's profile to the individual patient.**

### Patient & illness

Target symptoms (insomnia, low appetite). Prior response in the patient or a family member. Comorbid medical and psychiatric conditions.

### Drug factors

Side-effect profile. Drug interactions. Risk of overdose — the TCAs are lethal.

### Practical

Cost — newer agents can be expensive. Adherence and dosing convenience.

*The single safety point to carry: a suicidal patient should not be given a drug that is lethal in overdose.*

# SSRIs — first-line, best tolerated

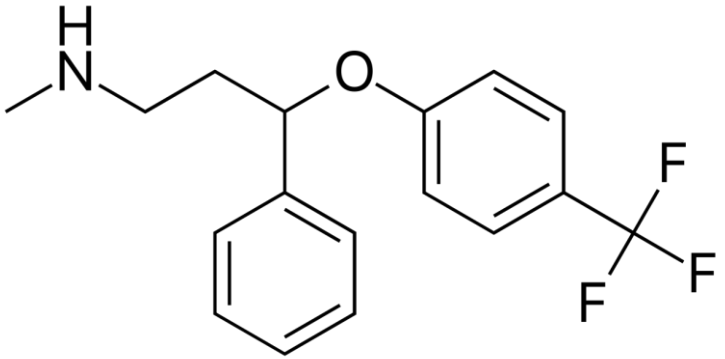
## First-line because they work and are safe in overdose.

### The agents

- Fluoxetine — long half-life, no taper needed
- Sertraline — fewest interactions; safe in cardiac disease
- Escitalopram and citalopram — most QT-prolonging
- Paroxetine — short half-life, worst discontinuation
- Fluvoxamine — CYP1A2 inhibitor, raises clozapine

### Common side-effects

- Anxiety, agitation, insomnia
- Nausea, diarrhoea, anorexia
- Sexual dysfunction — reduced libido, delayed ejaculation, anorgasmia




*Fluoxetine. Vaccinationist, Wikimedia Commons, public domain*

*Counsel patients that early jitteriness and GI upset usually settle within days — but sexual dysfunction tends to persist and is the commonest reason people stop. Ask about it directly.*

# SSRIs — serious adverse effects



## Know these

- Bleeding — impaired platelet 5-HT and aggregation; risk rises with NSAIDs and anticoagulants
- Hyponatraemia (SIADH) — especially the elderly; monitor $Na^+$, discontinue if < 125 mmol/L
- Serotonin syndrome
- Discontinuation syndrome

## Cardiac — QT

- Citalopram and escitalopram prolong QT the most
- Citalopram max 40 mg/day — 20 mg if > 60 y, hepatic impairment, or on CYP2C19 inhibitors
- Sertraline, fluoxetine, mirtazapine — safest, no meaningful QT effect
- Avoid TCAs in cardiac disease

*Check sodium in older patients starting an SSRI, and ask about NSAID or anticoagulant use before prescribing.*

# SNRIs — serotonin & noradrenaline



## Agents

- Venlafaxine and desvenlafaxine — dose-dependent hypertension
- Duloxetine — also for neuropathy and fibromyalgia; avoid in severe liver disease

## Clinical flags

- Venlafaxine has one of the worst discontinuation syndromes — short half-life
- Venlafaxine raises blood pressure at higher doses — check BP
- Dual action without the TCA receptor baggage

*SNRIs give dual serotonin–noradrenaline action without TCA receptor promiscuity, and duloxetine additionally helps neuropathic pain.*

# Tricyclic antidepressants



| Receptor blocked | Effects | Notes |
|---|---|---|
| **Antihistaminic** | Sedation, weight gain | Often the reason patients stop |
| **Anticholinergic** | Dry mouth, constipation, urinary retention, blurred vision | Worse in the elderly; avoid in prostatism and glaucoma |
| **Antiadrenergic ($\alpha_1$)** | Orthostatic hypotension | Falls risk |
| **Cardiac ($Na^+$ channel)** | QRS widening, QT prolongation, arrhythmia | LETHAL IN OVERDOSE — never prescribe freely to a suicidal patient |
| **Tertiary vs secondary** | Tertiary: imipramine, amitriptyline, doxepin, clomipramine — more side-effects | Secondary: desipramine, nortriptyline — metabolites of the tertiary amines, generally milder |

*Effective but toxic. Reserve TCAs, and remember they are the overdose-lethal class.*

# Overdose on the ward



A 22-year-old is brought to the ED two hours after taking a bottle of her mother's amitriptyline. She is drowsy, BP 88/54, heart rate 128.

The ECG shows a QRS of 130 ms and a tall R wave in aVR.

## Most important immediate treatment?

**A** Activated charcoal alone

**B** IV sodium bicarbonate

**C** IV flumazenil

**D** IV naloxone

*Commit to an answer before the next slide.*

# Overdose on the ward

ANSWER

A 22-year-old is brought to the ED two hours after taking a bottle of her mother's amitriptyline. She is drowsy, BP 88/54, heart rate 128.

The ECG shows a QRS of 130 ms and a tall R wave in aVR.

## Most important immediate treatment?

- **A** Activated charcoal alone
- **B** **IV sodium bicarbonate**
- **C** IV flumazenil
- **D** IV naloxone

### B — IV sodium bicarbonate.

TCA toxicity is sodium-channel blockade: QRS widening beyond 100 ms and a tall R in aVR predict seizures and arrhythmia. Sodium bicarbonate overcomes it. Charcoal alone does not treat the cardiotoxicity, and flumazenil is both wrong and dangerous here.

# Monoamine oxidase inhibitors



## Background & agents

- Iproniazid, an isoniazid derivative, was noted to lift mood in the 1950s
- Isocarboxazid, tranylcypromine, phenelzine; moclobemide is reversible
- Very effective — but reserved for atypical or treatment-resistant depression

## The two big dangers

- Hypertensive crisis with tyramine-rich foods — aged cheese, cured meats — or sympathomimetics
- Serotonin syndrome if combined with serotonergic drugs
- Washout needed: 2 weeks, or 5 weeks after fluoxetine
- Also orthostatic hypotension, weight gain, sexual dysfunction, insomnia



*The two exam points are the tyramine "cheese reaction" and the washout interval before or after a serotonergic drug.*

# Bupropion · mirtazapine · trazodone

**Learn these three by their signature — each is chosen for what it does besides lifting mood.**

## Bupropion (NDRI)

Activating. No sexual dysfunction and no weight gain, and it helps smoking cessation. But it LOWERS THE SEIZURE THRESHOLD — avoid in eating disorders and head injury — and it does not treat anxiety.

## Mirtazapine (NaSSA)

Sedating and increases appetite, which makes it useful where insomnia and poor intake dominate. Weight gain is the trade-off. No meaningful QT effect.

## Trazodone (SARI)

Used as a hypnotic. Daytime sedation is common, and it carries the rare but urgent risk of PRIAPISM — warn men to seek help immediately.

# Newer-generation antidepressants



## Serotonin-modulating

- Vortioxetine — multimodal serotonin modulator, pro-cognitive
- Vilazodone — serotonin partial-agonist reuptake inhibitor (SPARI)
- Agomelatine — melatonergic MT1/MT2 agonist plus 5-HT2C antagonist

## Rapid-acting / novel

- Esketamine, intranasal — NMDA antagonist for treatment-resistant depression and acute suicidality; now also monotherapy (2025)
- Dextromethorphan–bupropion (Auvelity, 2022) — rapid oral
- Zuranolone (2023) — oral, for postpartum depression

*The rapid-acting group matters for currency — all act far faster than the classic 4–6 weeks.*

# Agitated, sweating and twitchy



A 30-year-old on sertraline is given tramadol for back pain.

Within hours he becomes agitated and confused, sweating and febrile (39 °C), with a fast heart rate, brisk reflexes and clonus that is worse in the legs.

## Most likely diagnosis?

**A** Neuroleptic malignant syndrome

**B** Serotonin syndrome

**C** Anticholinergic toxicity

**D** Malignant hyperthermia

*Commit to an answer before the next slide.*

# Agitated, sweating and twitchy

ANSWER

A 30-year-old on sertraline is given tramadol for back pain.

Within hours he becomes agitated and confused, sweating and febrile (39 °C), with a fast heart rate, brisk reflexes and clonus that is worse in the legs.

## Most likely diagnosis?

**A** Neuroleptic malignant syndrome

**B** **Serotonin syndrome**

**C** Anticholinergic toxicity

**D** Malignant hyperthermia

### B — serotonin syndrome.

Rapid onset after adding a serotonergic drug, with the triad of mental-status change, autonomic instability and neuromuscular hyperactivity (clonus and hyperreflexia, worse in the legs). Stop the drugs, cool, give benzodiazepines, and cyproheptadine if severe.

# Serotonin syndrome

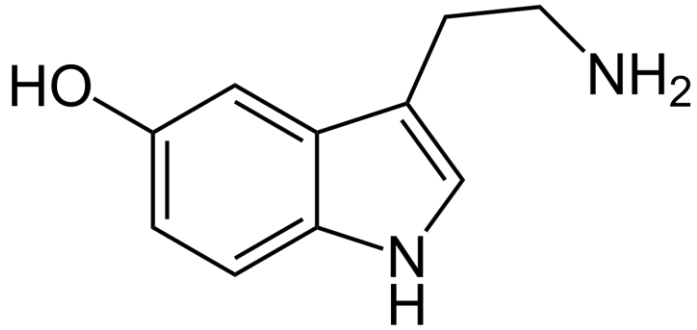
## A clinical diagnosis built on three pillars — and it comes on fast.

### The clinical triad

- Mental-status change — agitation, confusion
- Neuromuscular — hyperreflexia, clonus (especially lower limbs), myoclonus, ocular clonus, shivering, tremor
- Autonomic instability — tachycardia, hyperthermia, increased GI motility

### Culprits & management

- SSRIs/SNRIs with tramadol, meperidine, fentanyl; triptans; linezolid; MAOIs; ondansetron, metoclopramide; dextromethorphan
- Stop the offending agents; supportive care and cooling
- Benzodiazepines; cyproheptadine (5-HT2A antagonist) if severe




*Serotonin (5-HT). Jü, Wikimedia Commons, public domain*

*Tramadol, dextromethorphan, linezolid, triptans and St John's wort are the hidden serotonergic drugs that catch prescribers out.*

# Serotonin syndrome vs. NMS



| Feature | Serotonin syndrome | Neuroleptic malignant syndrome |
|---|---|---|
| **Cause** | Serotonergic drugs | Dopamine antagonists (or dopaminergic withdrawal) |
| **Onset** | Fast — within 24 hours | Slow — days to weeks |
| **Neuromuscular** | Hyperreflexia, clonus, myoclonus | "Lead-pipe" rigidity, bradyreflexia |
| **Shared** | Hyperthermia · autonomic instability · altered mental state · raised CK | Same |
| **Treatment** | Benzodiazepines + cyproheptadine | Stop drug + dantrolene / bromocriptine + benzodiazepines |

*Quick discriminator: reflexes up and fast onset = serotonin syndrome; rigidity and slow onset = NMS.*

# Antidepressant discontinuation syndrome



## FINISH

Not relapse and not addiction — a self-limiting withdrawal from stopping or missing an antidepressant. Worst with short half-life agents (paroxetine, venlafaxine). Onset within days of stopping; resolves in under 2 weeks. Taper to prevent it.

**F** **Flu-like**

Aches, lethargy, sweating

**I** **Insomnia**

With vivid dreams

**N** **Nausea**

With GI upset

**I** **Imbalance**

Dizziness, unsteadiness

**S** **Sensory**

Paraesthesiae, "brain zaps"

**H** **Hyperarousal**

Anxiety, irritability, agitation

*Distinguish from relapse: discontinuation starts within days and settles within two weeks; relapse builds over weeks and does not resolve on its own.*

# Stopped her tablets three days ago



A patient who has been well for eight months stops paroxetine abruptly when her prescription runs out.

Three days later she has dizziness, nausea, electric-shock sensations in her head and vivid dreams, and feels irritable. Her mood itself is no different from last week.

## Discontinuation syndrome, or relapse?

**A** Relapse of depression

**B** Discontinuation syndrome

**C** Serotonin syndrome

**D** A new anxiety disorder

*Commit to an answer before the next slide.*

# Stopped her tablets three days ago

ANSWER

A patient who has been well for eight months stops paroxetine abruptly when her prescription runs out.

Three days later she has dizziness, nausea, electric-shock sensations in her head and vivid dreams, and feels irritable. Her mood itself is no different from last week.

## Discontinuation syndrome, or relapse?

**A** Relapse of depression

**B** **Discontinuation syndrome**

**C** Serotonin syndrome

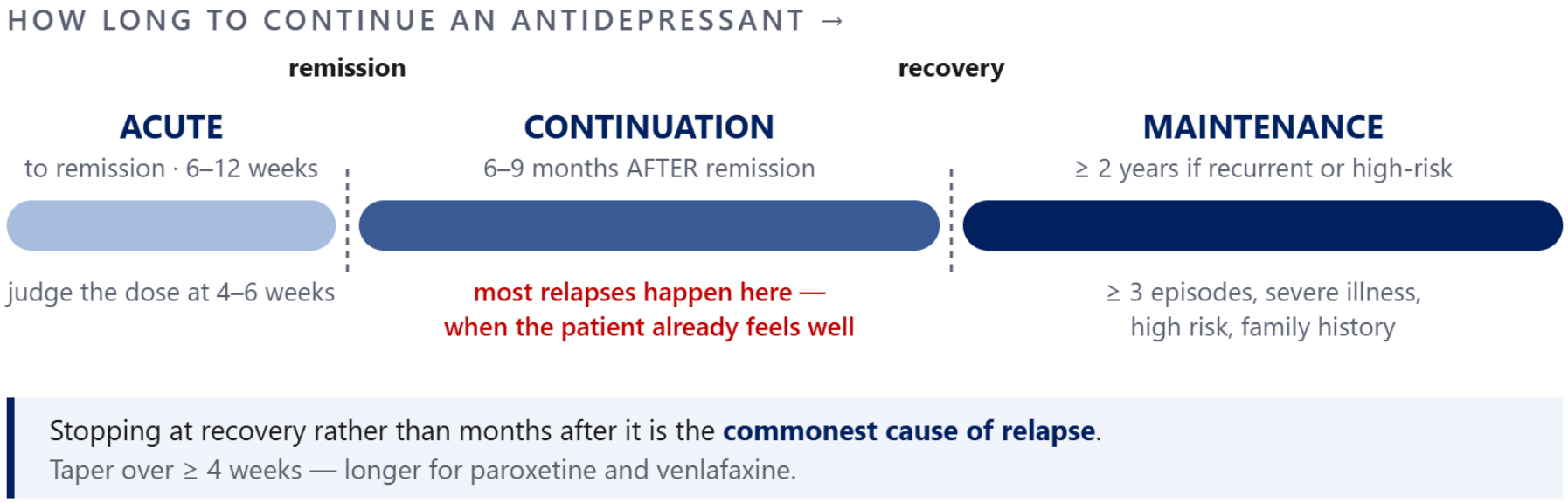
**D** A new anxiety disorder

### B — discontinuation syndrome.

Onset within days of stopping, dominated by physical symptoms, with mood unchanged. Relapse builds over weeks and leads with low mood and anhedonia. Restart the drug, then taper it properly — paroxetine's short half-life makes it one of the worst offenders.

# How long to continue treatment

## Choosing the drug is half the decision — the other half is how long to stay on it.




Stopping at recovery rather than months after it is the **commonest cause of relapse**.
Taper over ≥ 4 weeks — longer for paroxetine and venlafaxine.

*The other classes run to different clocks: benzodiazepines 2–4 weeks, not months; antipsychotics 1–2 years after a first episode of psychosis; lithium long-term, and never stopped abruptly — abrupt withdrawal can precipitate rebound mania.*

# Antidepressants in pregnancy & lactation



**Untreated severe depression is itself a risk to mother and baby — blanket stopping is often the wrong call.**

## Pregnancy

- SSRIs are not major teratogens; no robust link to cardiac anomalies after adjusting for confounders
- Paroxetine — historic, weak cardiac signal; generally avoided in the first trimester
- Late use may cause transient neonatal adaptation; the decision balances relapse risk

## Lactation

- Overall infant risk is low
- Lowest infant levels: sertraline, paroxetine
- Higher: citalopram, fluoxetine, venlafaxine



*Women with significant or recurrent illness are usually not good candidates for stopping medication.*

BMC
Batterjee Medical College

PART 2

# Anxiolytics & Sedatives

Benzodiazepines — mechanism, uses, risks, tolerance and withdrawal — plus the Z-drugs and buspirone.

# Benzodiazepines — mechanism & uses

**"Benzos = Frequency; Barbiturates = Duration" — a classic exam distinction.**

## Mechanism (high-yield)

- Bind the GABA-A receptor and increase the FREQUENCY of chloride-channel opening
- Contrast: barbiturates increase the DURATION of opening — which is why they are lethal alone
- The frequency/duration split explains the difference in overdose safety

## Uses

- Anxiety; short-term insomnia
- Muscle relaxation
- Anticonvulsant — status epilepticus
- Rapid tranquillisation
- Alcohol and sedative withdrawal

# Choosing a benzodiazepine & reversing it



| Group | Agents | Behaviour / why it matters |
|---|---|---|
| **Short-acting** | Alprazolam, lorazepam, oxazepam, temazepam | More rebound and withdrawal between doses |
| **Long-acting** | Diazepam, chlordiazepoxide, clonazepam | Smoother cover, but they accumulate — caution in the elderly |
| **"LOT" — liver-safe** | Lorazepam · Oxazepam · Temazepam | Phase-II glucuronidation only, no oxidative metabolism — preferred in liver disease and the elderly |
| **Reversal** | Flumazenil | Reverses overdose but CAN PRECIPITATE SEIZURES in dependence or mixed overdose — used rarely; supportive airway care is the mainstay |

*Half-life predicts behaviour; the LOT drugs are the answer whenever the liver is compromised.*

# Side-effects, tolerance & withdrawal



## Side-effects & misuse

- Sedation, drowsiness, cognitive and psychomotor impairment, falls in the elderly
- Anterograde amnesia; paradoxical disinhibition
- Dependence — tolerance can develop within ~2 weeks

## Withdrawal — can be dangerous

- Increased anxiety, sweats, tremor, hypersensitivity, aches, headache
- SEIZURES, COMA, DEATH — like alcohol withdrawal
- Never stop abruptly in a dependent patient — taper gradually

*Pregnancy: avoid if possible — cleft palate signal, floppy-infant syndrome and neonatal withdrawal if used near term; present in breast milk.*

# Day 3 after his hernia repair



A 58-year-old is admitted for elective hernia repair. He has taken diazepam 20 mg daily for years; it was not written up on his admission drug chart.

On day 3 he is tremulous and sweating with a BP of 175/100, and then has a generalised tonic-clonic seizure. He does not drink alcohol.

## What has happened, and what does he need?

**A** Alcohol withdrawal → thiamine

**B** Benzodiazepine withdrawal → reinstate a benzodiazepine

**C** Post-operative delirium → an antipsychotic

**D** Serotonin syndrome → cyproheptadine

*Commit to an answer before the next slide.*

# Day 3 after his hernia repair

ANSWER

A 58-year-old is admitted for elective hernia repair. He has taken diazepam 20 mg daily for years; it was not written up on his admission drug chart.

On day 3 he is tremulous and sweating with a BP of 175/100, and then has a generalised tonic-clonic seizure. He does not drink alcohol.

## What has happened, and what does he need?

**A** Alcohol withdrawal → thiamine

**B** **Benzodiazepine withdrawal → reinstate a benzodiazepine**

**C** Post-operative delirium → an antipsychotic

**D** Serotonin syndrome → cyproheptadine

### B — benzodiazepine withdrawal. Reinstate, then taper.

Omitting a long-standing benzodiazepine on admission causes a withdrawal that looks exactly like alcohol withdrawal — tremor, sweating, hypertension, and seizures that can be fatal. Reinstate the benzodiazepine, then plan a deliberate taper. Reconcile psychotropics on every admission.

# Z-drugs & buspirone



## Z-drugs (hypnotics)

- Zolpidem, zopiclone, zaleplon
- Act at the GABA-A benzodiazepine site, more selectively
- For short-term insomnia — but still cause dependence, complex sleep behaviours and falls

## Buspirone (anxiolytic)

- 5-HT1A partial agonist
- No dependence, no sedation, no withdrawal
- Delayed onset of 1–2 weeks — useless for acute anxiety; an adjunct in GAD

*Buspirone is the outlier: the one anxiolytic with no dependence — at the cost of being no use in a crisis.*

BMC
Batterjee Medical College

PAUSE

# Break — 10 minutes

After the break: antipsychotics and the four dopamine pathways · the EPS timeline · NMS · clozapine · mood stabilisers and lithium · then the integration cases.

BMC
Batterjee Medical College

PART 3

# Antipsychotics

Typical vs atypical · the side-effect map · EPS and tardive dyskinesia · NMS · metabolic and prolactin effects · clozapine · special populations.

# Fever, rigidity and confusion



A 24-year-old started on haloperidol 5 days ago develops a high fever, generalised 'lead-pipe' muscle rigidity, fluctuating consciousness, labile blood pressure and tachycardia.

Bloods show a markedly raised creatine kinase.

## Most likely diagnosis?

**A** Serotonin syndrome

**B** Neuroleptic malignant syndrome

**C** Malignant catatonia

**D** Meningitis

*Commit to an answer before the next slide.*

# Fever, rigidity and confusion



A 24-year-old started on haloperidol 5 days ago develops a high fever, generalised 'lead-pipe' muscle rigidity, fluctuating consciousness, labile blood pressure and tachycardia.

Bloods show a markedly raised creatine kinase.

## Most likely diagnosis?

**A** Serotonin syndrome

**B** **Neuroleptic malignant syndrome**

**C** Malignant catatonia

**D** Meningitis

## B — neuroleptic malignant syndrome.

Days after a dopamine antagonist: fever, 'lead-pipe' rigidity, autonomic instability, altered consciousness and a raised CK. Stop the antipsychotic; supportive care; dantrolene / bromocriptine and benzodiazepines.

# Mechanism — four dopamine pathways

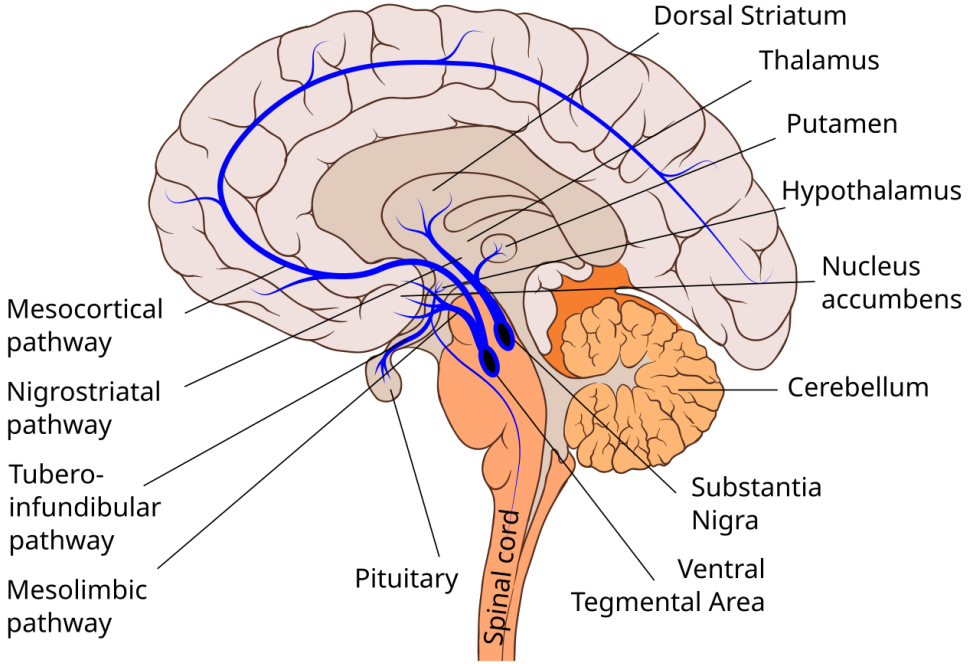
## One diagram explains the whole antipsychotic side-effect story.

### What D2 blockade does in each

- Mesolimbic → the therapeutic target, reduces positive symptoms
- Mesocortical → blockade may WORSEN negative and cognitive symptoms
- Nigrostriatal → extrapyramidal side-effects
- Tuberoinfundibular → hyperprolactinaemia

### Typical vs atypical

- Typical (FGA): mainly D2 antagonists
- Atypical (SGA): 5-HT2A + D2 antagonism → fewer EPS, some effect on negative symptoms, more metabolic effects




*Dopaminergic pathways. Slashme / Patrick J. Lynch, Wikimedia Commons, CC BY-SA 4.0*

*Atypicals add 5-HT2A antagonism, which lifts the striatal brake — fewer movement disorders, at the cost of weight and metabolic risk.*

# Typical antipsychotics (FGA)



## Agents & action

- Chlorpromazine, fluphenazine, haloperidol, perphenazine
- Primarily D2 antagonists
- Treat positive symptoms; little effect on negative symptoms

## High vs low potency

- High-potency (haloperidol, fluphenazine) → more EPS and prolactin
- Low-potency (chlorpromazine) → more anticholinergic effects, sedation and hypotension
- Key risks across the class: EPS, tardive dyskinesia, NMS

*Potency is the useful split — it predicts which side-effects the patient will get.*

# Atypical antipsychotics (SGA)



**Selection is about matching side-effect profile to the patient — not a blanket "atypicals are better".**

## Agents & action

- Risperidone, paliperidone, olanzapine, quetiapine, ziprasidone, lurasidone, asenapine, clozapine
- 5-HT2A + D2 antagonism
- Aripiprazole, brexpiprazole and cariprazine are D2 PARTIAL AGONISTS — a different mechanism, often called third-generation

## Practical

- No clear efficacy advantage over FGAs — except clozapine
- Fewer EPS but higher metabolic risk
- Lurasidone and cariprazine are also licensed for BIPOLAR DEPRESSION
- Judge efficacy only after ≥ 4 weeks at an adequate dose

# Antipsychotics — the side-effect map

## Across a ROW = choosing a drug. Down a COLUMN = explaining a symptom.

| | Sedation | Weight | EPS | Anti-Ch | ↓ BP | Prolactin | QTc |
|---|---|---|---|---|---|---|---|
| **FIRST-GENERATION (D2 antagonists)** | | | | | | | |
| **Haloperidol** | ● | ● | ●●● | — | ● | ●●● | ●● |
| **Chlorpromazine** | ●●● | ●● | ●● | ●● | ●●● | ●●● | ● |
| **SECOND-GENERATION (5-HT2A + D2 antagonists)** | | | | | | | |
| **Clozapine** | ●●● | ●●● | — | ●●● | ●●● | — | ● |
| **Olanzapine** | ●● | ●●● | ● | ● | ● | ● | — |
| **Quetiapine** | ●●● | ●● | — | ● | ●● | — | ● |
| **Risperidone** | ● | ●● | ●● | — | ●● | ●●● | ● |
| **Ziprasidone** | ● | — | ● | — | ● | ● | ●●● |
| **Lurasidone** | ● | — | ●● | — | — | ● | — |
| **D2 PARTIAL AGONISTS ("third-generation")** | | | | | | | |
| **Aripiprazole** | — | ● | ●● | — | — | — | — |
| **Brexpiprazole** | ● | ●● | ● | — | ● | — | — |
| **Cariprazine** | — | — | ●● | — | — | — | — |

— none / minimal  ● mild  ●● moderate  ●●● marked

*A teaching ranking, not a measured scale. Partial-agonist EPS is mainly AKATHISIA — most with cariprazine, least with brexpiprazole.*

# A young man with a twisted neck



A 19-year-old is given haloperidol in the ED for agitation.

A few hours later his neck twists to one side and his eyes roll upward and lock in place; he is frightened but fully conscious.

## Diagnosis & treatment?

**A** Tardive dyskinesia → stop drug

**B** Acute dystonia → benztropine

**C** Akathisia → propranolol

**D** Seizure → lorazepam

*Commit to an answer before the next slide.*

# A young man with a twisted neck



A 19-year-old is given haloperidol in the ED for agitation.

A few hours later his neck twists to one side and his eyes roll upward and lock in place; he is frightened but fully conscious.

## Diagnosis & treatment?

**A** Tardive dyskinesia → stop drug

**B** **Acute dystonia → benztropine**

**C** Akathisia → propranolol

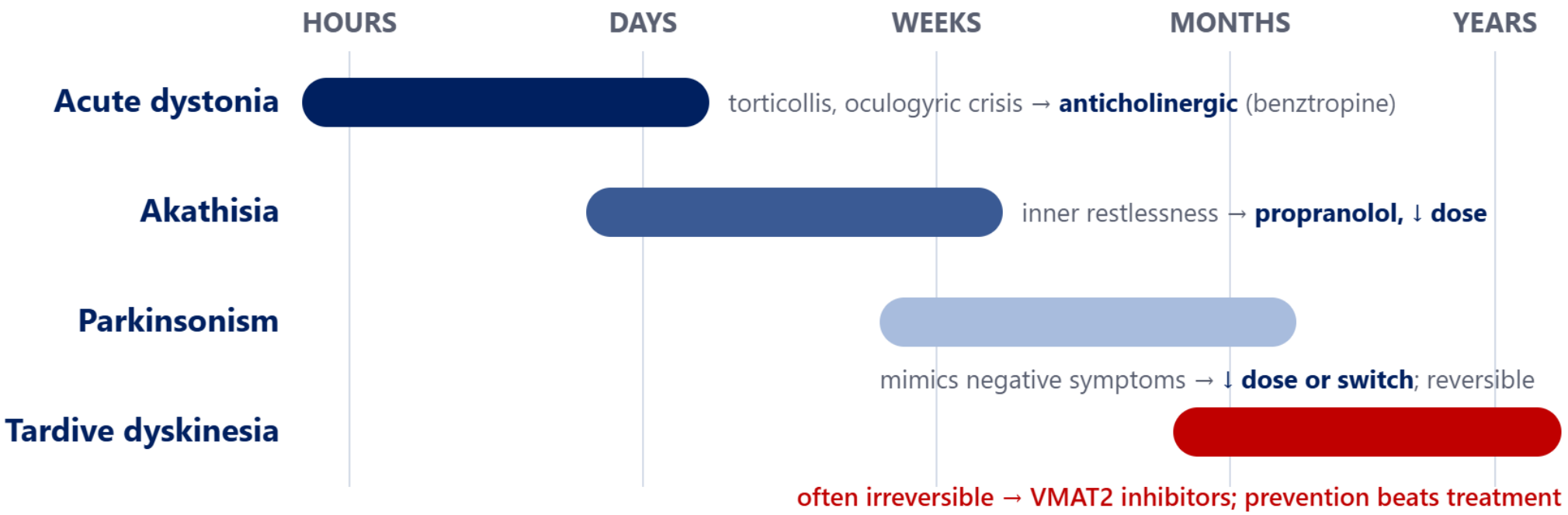
**D** Seizure → lorazepam

**B — acute dystonia → anticholinergic (benztropine or diphenhydramine).**

Hours after a high-potency antipsychotic: sustained muscle spasm — torticollis and oculogyric crisis. It responds rapidly to an anticholinergic. Young men are at highest risk.

# Extrapyramidal symptoms — timeline






> The trap: anticholinergics help dystonia and parkinsonism, do little for akathisia, and **WORSEN tardive dyskinesia**.

*The trap: anticholinergics help dystonia and parkinsonism, do little for akathisia, and WORSEN tardive dyskinesia. Highest TD risk with long-term high-potency FGAs.*

# Acute dystonia — what you actually see



**The two signs from the vignette, made visible: the eyes locked upward and the neck and trunk thrown into spasm — hours after a high-potency antipsychotic, and fully conscious throughout.**

**THE EYES**



## Oculogyric crisis

Sustained conjugate UPWARD deviation of both eyes

**THE NECK & TRUNK**



## Acute dystonic reaction

Torticollis & truncal spasm · dystonic arm posturing

*Preserved consciousness separates acute dystonia from a seizure; it reverses in minutes with an IM/IV anticholinergic. Eyes: Tokioka K et al., Clin Case Rep 2026, CC BY 4.0. Posture: James Heilman MD, Wikimedia Commons, CC BY-SA 3.0.*

# Neuroleptic malignant syndrome (NMS)



## Presentation

- Fever, 'lead-pipe' rigidity, autonomic instability, fluctuating consciousness
- Raised CK, leukocytosis, altered LFTs
- Myoglobinuria → renal failure

## Risk & management

- Risk: high-potency FGAs, rapid dose increase, dehydration, agitation, young males
- Stop the antipsychotic; monitor, rehydrate, cool
- Benzodiazepines; dantrolene + bromocriptine (± amantadine or L-dopa)
- Hold antipsychotics ≥ 5 days; reintroduce cautiously later

*NMS builds over days — contrast with the hours of serotonin syndrome.*

# Weight gain & metabolic syndrome

## Driven by H1, 5-HT2C and D2 antagonism, plus leptin changes — the price of the atypicals.

| Relative risk | Agents | What to do |
|---|---|---|
| **High** | Clozapine, olanzapine | Expect significant weight gain — monitor closely from baseline |
| **Moderate** | Quetiapine, risperidone, brexpiprazole | Monitor; consider switching if weight climbs |
| **Low / neutral** | Aripiprazole, cariprazine, ziprasidone, lurasidone | Preferred where metabolic risk already exists |
| **Monitoring** | All antipsychotics | Baseline; weight weekly for 6 weeks; weight, waist, glucose/HbA1c, lipids and BP at 12 weeks; then at 1 year and annually |
| **Management** | — | Switch to a weight-neutral agent; consider metformin; lifestyle support; refer for cardiometabolic care |

*Relative risk categories as taught; this is a ranking, not a measured scale.*

# Hyperprolactinaemia

## Dopamine normally inhibits prolactin — so D2 antagonists raise it.

### What it causes

- Galactorrhoea, amenorrhoea, gynaecomastia
- Sexual dysfunction
- Long-term: osteoporosis
- Worst: risperidone, paliperidone, amisulpride, haloperidol

### Prolactin-sparing & cautions

- Sparing: aripiprazole, brexpiprazole, cariprazine, quetiapine, clozapine (olanzapine low)
- Avoid prolactin-raising agents under 25 years (peak bone mass), in osteoporosis, and in hormone-dependent breast cancer
- Manage by switching, reducing dose, or adding aripiprazole

# QT prolongation & other effects



## QT prolongation

- Antipsychotics block cardiac potassium channels and prolong the QT interval
- Usually trivial — but beyond 500 ms the risk of torsades de pointes climbs
- Check an ECG and electrolytes in at-risk patients
- Higher risk: haloperidol (especially IV), ziprasidone, pimozide

## Also watch for

- Anticholinergic effects
- Seizures — dose-related, chlorpromazine and clozapine
- Orthostatic hypotension
- Hyponatraemia (SIADH)

# Clozapine — most effective, most monitored

**The paradox of psychiatry: the single most effective antipsychotic, and the most dangerous.**

## Why it's special

- Most effective agent — reserved for treatment-resistant schizophrenia after ≥ 2 failed trials
- The ONLY antipsychotic that reduces suicidality
- Very low EPS and prolactin

## The serious risks — monitor

- Agranulocytosis (~1%) → mandatory ANC monitoring: weekly × 6 months, then 2-weekly × 6 months, then monthly
- ANC < 1.5 $\times 10^9$/L → withhold and repeat; < 0.5 $\times 10^9$/L → agranulocytosis, stop permanently
- Myocarditis — early weeks
- Seizures — dose-related
- Constipation → ileus, which can be fatal; sialorrhoea; sedation; weight gain



*Constipation is the risk students forget — clozapine-induced ileus kills more patients than agranulocytosis.*

# Sore throat on clozapine



A 29-year-old who has been on clozapine for four months attends the clinic with a temperature of 38.6 °C and a sore throat that started yesterday.

He looks otherwise well and his chest is clear. He asks whether he really needs to be seen today.

## What is your first action?

**A** Reassure — it is probably viral

**B** Urgent FBC and absolute neutrophil count today

**C** Prescribe antibiotics and review in a week

**D** Stop clozapine and discharge him

*Commit to an answer before the next slide.*

# Sore throat on clozapine

ANSWER

A 29-year-old who has been on clozapine for four months attends the clinic with a temperature of 38.6 °C and a sore throat that started yesterday.

He looks otherwise well and his chest is clear. He asks whether he really needs to be seen today.

## What is your first action?

**A** Reassure — it is probably viral

**B** **Urgent FBC and absolute neutrophil count today**

**C** Prescribe antibiotics and review in a week

**D** Stop clozapine and discharge him

## B — urgent FBC and ANC, today.

Fever or sore throat in a patient on clozapine is agranulocytosis until proven otherwise. Check the count the same day and withhold clozapine until you have it: ANC below 1.5 $\times 10^9$/L means withhold and repeat, below 0.5 $\times 10^9$/L is agranulocytosis and clozapine is stopped for good. Reassurance and a week's wait can be fatal.

# Long-acting injectables & special populations



## Long-acting injectables

FGA: haloperidol, fluphenazine decanoate. SGA: risperidone, paliperidone, aripiprazole, olanzapine. They transform adherence in relapsing illness.

## Dementia — black box

Increased mortality in elderly patients with dementia (≈2.6% vs ≈4.5% over 10-week trials), from cardiovascular events and falls. Use only if essential, at the lowest dose.

## Renal & liver

Renal: avoid sulpiride/amisulpride and strong anticholinergics; consider haloperidol or low-dose olanzapine. Liver: monitor LFTs — transaminitis in the first 1–6 weeks.

BMC
Batterjee Medical College

PART 4

# Mood Stabilisers

Lithium and its toxicity · valproate · carbamazepine · lamotrigine — indications, monitoring and the dangerous interactions.

# Tremor and unsteadiness on lithium



A 45-year-old on long-term lithium starts ibuprofen for back pain and, during a hot week, develops a coarse tremor, unsteady gait, vomiting and slurred speech.

She seems confused.

## What is happening?

**A** Lithium toxicity

**B** Serotonin syndrome

**C** Hypoglycaemia

**D** Cerebellar stroke

*Commit to an answer before the next slide.*

# Tremor and unsteadiness on lithium



A 45-year-old on long-term lithium starts ibuprofen for back pain and, during a hot week, develops a coarse tremor, unsteady gait, vomiting and slurred speech.

She seems confused.

## What is happening?

**A** **Lithium toxicity**

**B** Serotonin syndrome

**C** Hypoglycaemia

**D** Cerebellar stroke

## A — lithium toxicity.

NSAIDs, dehydration, thiazides and ACE-inhibitors reduce lithium clearance and raise the level. Coarse tremor → ataxia → confusion signals toxicity. Hold lithium, hydrate, check the level; dialyse if severe.

# Lithium — dosing & monitoring

## Gold standard, and the only mood stabiliser that clearly reduces suicide — but it demands respect.

### Dosing & levels

- ~900–1800 mg/day, divided
- Therapeutic 0.6–1.2 mEq/L; maintenance 0.6–0.8
- 1.2–1.5 is ABOVE RANGE — recheck the level and review the dose; toxic above 1.5
- Monitor: renal function, TSH, Ca/PTH, ECG, and the level

### Side-effects

- Thyroid abnormalities; benign leukocytosis
- Polyuria and polydipsia — nephrogenic diabetes insipidus
- Weight gain; fine tremor; cognitive slowing
- Acne, hair loss
- Teratogenic — Ebstein's anomaly (small risk)



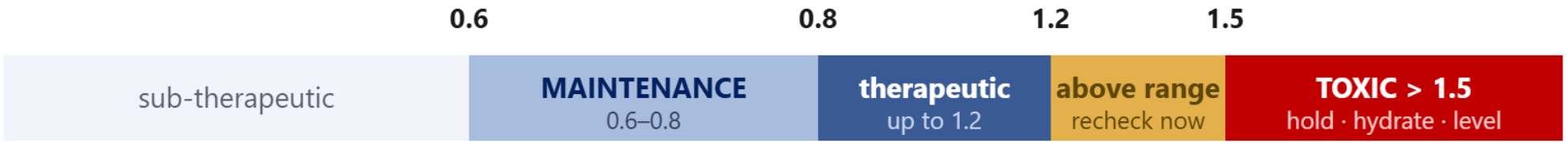
*Lithium carbonate. Wikimedia Commons, public domain*

*Narrow therapeutic index — see the next slide for what tips a stable patient over.*

# Lithium toxicity & interactions



SERUM LITHIUM (mEq/L) — A NARROW THERAPEUTIC INDEX




**Toxicity by stage**

**Early** — lethargy, COARSE tremor, vomiting, diarrhoea

**Moderate** — confusion, dysarthria, nystagmus, ECG changes

**Late** — seizures, hyperreflexia, renal failure, coma; **dialyse if severe**

**What tips a stable patient over**

dehydration · NSAIDs · thiazides

ACE-inhibitors · renal impairment

Monitor: the level, renal function, TSH, Ca/PTH, ECG. Never stop abruptly — rebound mania.

*The therapeutic index is narrow — toxicity begins just above the therapeutic range, so any new drug or dehydrating illness can tip a stable patient over.*

# Valproate



**Best for acute mania and mixed or rapid-cycling states — and the most teratogenic mood stabiliser.**

## Side-effects

- Thrombocytopenia and platelet dysfunction
- Nausea, weight gain, transaminitis (hepatotoxicity)
- Sedation, tremor, hair loss
- Pancreatitis; hyperammonaemia

## Pregnancy — avoid

- Neural-tube defects: 1–2% vs 0.14–0.2% background — folate-related
- Lower childhood IQ
- Avoid in women of childbearing potential wherever possible

# Carbamazepine



## Side-effects

- Rash — the most common
- GI upset, transaminitis
- Sedation, dizziness, ataxia, diplopia
- AV conduction delay; hyponatraemia (SIADH-like)
- Aplastic anaemia / agranulocytosis — rare

## High-yield

- HLA-B*1502 → carbamazepine-induced SJS/TEN in Han Chinese and South-East Asian patients — GENOTYPE BEFORE STARTING
- Auto-induction of CYP3A4 — its own level falls over the first weeks
- A strong enzyme inducer, so many interactions

*The pharmacogenomic pearl — HLA-B*1502 — is the single most examinable fact about this drug.*

# Lamotrigine



## Profile

- Best for bipolar DEPRESSION and maintenance
- Weak for acute mania
- Generally well tolerated; no routine level monitoring

## The rash rule

- A simple rash can progress to Stevens–Johnson syndrome or TEN
- Titrate slowly to reduce the risk
- VALPROATE DOUBLES LAMOTRIGINE LEVELS by inhibiting glucuronidation → use a slower titration and a lower dose

*Any rash on lamotrigine is stopped and assessed — do not "watch and wait".*

# The lamotrigine rash — three severities

**The same drug reaction, graded. The first is common and benign; the last is a dermatological emergency.**

**COMMON — BENIGN**



## Morbilliform rash

Skin only. No mucosa, no blisters, no systemic illness

**RARE — EMERGENCY**



## Stevens–Johnson syndrome

MUCOSAL involvement · detachment < 10% of body surface

**LIFE-THREATENING**



## Toxic epidermal necrolysis

Sheet-like loss > 30% of body surface · mortality ~25–35%

*Photographs illustrate the reaction patterns, not lamotrigine cases. 1: Frank C. Müller, CC BY-SA 2.5 · 2: James Heilman MD, CC BY-SA 3.0 · 3: AfroBrazilian, CC BY-SA 3.0 — all Wikimedia Commons.*

BMC
Batterjee Medical College

PART 5

# Integration & Review

Cytochrome-P450 interactions · black-box warnings · board questions · and sources.

# Cytochrome-P450 interactions

**Sertraline, escitalopram and mirtazapine have the fewest CYP interactions — useful in the medically complex or polypharmacy patient.**

## Inhibitors — raise other drug levels

- Fluoxetine, paroxetine → CYP2D6: raise TCAs and some antipsychotics; codeine becomes ineffective
- Fluvoxamine → CYP1A2: raises clozapine and theophylline
- Watch serotonergic combinations → serotonin syndrome

## Inducers — lower other drug levels

- Carbamazepine — potent inducer: lowers the oral contraceptive, lamotrigine and many drugs; auto-induces
- Smoking induces CYP1A2 → lowers clozapine and olanzapine
- If the patient QUITS smoking, those levels RISE — a classic trap

# Psychotropics on a non-psychiatric ward



**You will rarely start these drugs. You will constantly look after patients already taking them — and that is where the harm happens.**

## Reconcile on admission

Psychotropics are among the most commonly omitted drugs on an admission chart, and the most dangerous to omit. Check what the patient actually takes, and write it up.

## Never stop abruptly

Benzodiazepines → withdrawal seizures. Antidepressants → discontinuation syndrome. Clozapine → more than ~48 hours off means re-titrating from a low dose.

## Nil by mouth & dehydrated

The classic lithium-toxicity setting: surgery, vomiting, diarrhoea, hot weather, or a new NSAID / thiazide / ACE-inhibitor. Hold the lithium, rehydrate, check the level.

## Fever on clozapine

FBC and ANC the same day — agranulocytosis until proven otherwise. And take constipation seriously: clozapine ileus kills more patients than agranulocytosis does.



*If you remember one slide from this lecture as a future surgeon or physician rather than a psychiatrist, remember this one.*

# The interactions you will cause

**These are not exotic psychiatric drugs — they are the everyday drugs YOU will prescribe, colliding with the psychotropic the patient is already on.**

| What you prescribe | Hits | What happens | What to do |
|---|---|---|---|
| **Tramadol** | SSRI / SNRI | Serotonin syndrome — the commonest trigger in practice | Choose a non-serotonergic analgesic |
| **Codeine** | Fluoxetine, paroxetine | CYP2D6 inhibition blocks conversion to morphine — NO analgesia | Use a different analgesic; codeine will simply not work |
| **NSAIDs** | Lithium / SSRI | Lithium level rises → toxicity. With SSRIs, GI bleeding risk rises | Avoid; if unavoidable, check the level and add gastric protection |
| **Clarithromycin, linezolid** | Clozapine / serotonergics | Enzyme inhibition raises clozapine; linezolid is an MAOI → serotonin syndrome | Check interactions before prescribing an antibiotic |
| **Any QT-prolonging drug** | Antipsychotics | Additive QT prolongation → torsades | ECG and electrolytes before and after |

*Also: metoclopramide and ondansetron are serotonergic, and metoclopramide is a dopamine antagonist that causes EPS and can contribute to NMS.*

# Black-box warnings to know



| Drug / class | Black-box warning |
|---|---|
| **All antidepressants** | Increased suicidal ideation and behaviour in patients under 25, early in treatment |
| **Antipsychotics (all)** | Increased mortality in elderly patients with dementia-related psychosis |
| **Clozapine** | Agranulocytosis · myocarditis · seizures · orthostatic hypotension and syncope |
| **Valproate** | Hepatotoxicity · pancreatitis · teratogenicity (neural-tube defects) |
| **Carbamazepine** | Aplastic anaemia / agranulocytosis · SJS/TEN (HLA-B*1502) |

*These shape both prescribing and consent — they are what you must tell the patient.*

## Mr. H, 34 — restless and shivering in ED

Mr. H, 34, is brought to the emergency department by his brother, who says he became "jittery and strange" over the course of the afternoon.

He has taken paroxetine for depression for eight months. Two days ago a walk-in clinic gave him tramadol for back pain, and he has also been taking an over-the-counter cough syrup containing dextromethorphan.

On examination he is anxious and agitated, drenched in sweat, pulse 124, blood pressure 158/92, temperature 38.4 °C. His pupils are dilated, bowel sounds are hyperactive, and he has tremor with clonus at the ankles that is more marked in the legs than the arms. Reflexes are brisk.

He is not rigid in the neck or jaw. He takes no antipsychotic. Symptoms began within hours, not days.

# Case problem — work through it



## Questions

- 1 · What is the diagnosis, and which drugs caused it?
- 2 · How do you distinguish it from NMS at the bedside?
- 3 · What is the immediate management?
- 4 · Which antidote applies here, and which applies to NMS?
- 5 · What must the discharge prescription and advice cover?

## Model answer

- 1 · Serotonin syndrome — paroxetine + tramadol + dextromethorphan; the last two are the agents most often overlooked
- 2 · Onset hours vs days · clonus and hyperreflexia worse in the legs vs lead-pipe rigidity · dilated pupils · hyperactive vs reduced bowel sounds · serotonergic vs antipsychotic
- 3 · Stop every serotonergic drug. IV fluids, active cooling, cardiac monitoring, benzodiazepines for agitation
- 4 · Cyproheptadine for serotonin syndrome; dantrolene and bromocriptine for NMS — do not swap them
- 5 · Document as a drug interaction, warn that tramadol and dextromethorphan are serotonergic, plan the washout before any new serotonergic agent

*Finish by asking what would change if he had instead been started on haloperidol three days ago and now had rigidity with quiet bowel sounds.*

## Ms. K, 27 — "she's just constipated"

Ms. K, 27, has taken clozapine 400 mg daily for two years for treatment-resistant schizophrenia. Her illness has been stable on it after two previous antipsychotics failed.

Her mother brings her to the ED because she has not opened her bowels for six days. Over the last two days she has become distended and has begun vomiting. She was started on procyclidine three weeks ago for tremor.

On examination her abdomen is distended and tympanic with almost no bowel sounds. Temperature 37.4 °C, pulse 104. Her ANC last week was normal. She is not rigid, and she is fully oriented.

The surgical registrar asks whether her clozapine can simply be stopped while she is investigated.

# Case problem — work through it



## Questions

- 1 · What is the diagnosis, and what made it worse?
- 2 · Why is this more dangerous than students expect?
- 3 · How would you manage her now?
- 4 · Should the clozapine be stopped, as the registrar suggests?
- 5 · What should have prevented this?

## Model answer

- 1 · Clozapine-induced GI hypomotility progressing to ileus, aggravated by the anticholinergic procyclidine
- 2 · Commoner than agranulocytosis and with higher mortality — but no mandatory monitoring, so it is missed
- 3 · Surgical assessment and imaging, stop the anticholinergic, aggressive laxatives, nil by mouth if obstructed; treat as a surgical emergency
- 4 · Not casually. Stopping > 48 hours means re-titrating from a low dose, and she has already failed two other antipsychotics — decide jointly with psychiatry
- 5 · Asking about bowels at every clozapine review, and prophylactic laxatives rather than adding an anticholinergic

*The transferable lesson: monitoring protocols create blind spots. The risk that is counted gets watched; the risk that is not counted gets missed.*

# Board-style questions



| # | Question | Answer |
|---|---|---|
| 1 | Fast onset, clonus and hyperreflexia after adding tramadol to an SSRI? | Serotonin syndrome → benzodiazepines + cyproheptadine |
| 2 | 'Lead-pipe' rigidity, fever and raised CK days after starting haloperidol? | NMS → stop drug, dantrolene / bromocriptine |
| 3 | Which antidepressant to avoid in a patient with an eating disorder? | Bupropion — it lowers the seizure threshold |
| 4 | Which benzodiazepines are safe in liver disease? | Lorazepam, Oxazepam, Temazepam — "LOT" |
| 5 | Genotype to check before carbamazepine in a Han Chinese or South-East Asian patient? | HLA-B*1502 — SJS/TEN risk |

*If you can answer these on sight, you have the exam-critical core.*

# Summary — seven things to remember



## Golden rules

- Antidepressants: similar efficacy, differ in side-effects; ~60–70% respond at 4–6 weeks
- Serotonin syndrome (reflexes up, fast) vs NMS (rigidity, slow)
- Benzos = frequency; barbiturates = duration; "LOT" safe in liver disease

## …continued

- EPS timeline: dystonia → akathisia → parkinsonism → tardive dyskinesia (VMAT2 inhibitors)
- Clozapine = most effective and anti-suicidal — but monitor the ANC
- Lithium: level 0.6–1.2, toxic > 1.5; NSAIDs, thiazides and ACE-inhibitors raise it
- Carbamazepine → HLA-B*1502; lamotrigine → SJS (slow titration; valproate raises the level)

# References

## Facts and figures cross-checked against current guidelines.

### Core references

- Maudsley Prescribing Guidelines in Psychiatry, 14th ed.
- Kaplan & Sadock's Synopsis of Psychiatry
- Stahl's Essential Psychopharmacology
- NICE guidance — depression, bipolar disorder, psychosis

### Regulatory & updates

- FDA labelling and black-box warnings
- Citalopram QT (2011) · VMAT2 inhibitors (2017)
- Auvelity (2022) · zuranolone (2023) · esketamine monotherapy (2025)
- FDA pharmacogenomics — HLA-B*1502 and carbamazepine



*Image and icon credits: _assets\images\ATTRIBUTIONS.md.*

BMC
Batterjee Medical College

# Thank You

Dr. Hussain M. Khrad, MBBS, ABNP, ABAM, MA · Department of Psychiatry

Questions welcome. Sources for every figure are on the references slide.